

EXECUTIVE SUMMARY

Factory Reallocation & Shipping Optimization Recommendation System

Nassau Candy Distributor · Unified Mentor Internship · June 2026

This document provides a concise summary of findings and recommendations from the Factory Reallocation & Shipping Optimization project, prepared for Nassau Candy Distributor's leadership team. The full technical analysis, dataset descriptions, and model documentation are available in the accompanying Research Paper.

THE SITUATION

Why Action Is Required

Nassau Candy Distributor's five factories — spanning Arizona to Minnesota — have not had their product assignments reviewed systematically in years. The result is a supply chain that assigns products to factories by tradition rather than by performance, creating three quantifiable problems:

- Excess Lead Times:** Sugar Shack (MN) ships 5 products, including to West-coast customers — incurring lead times of 6.2-7.1 days, vs. 3.4-4.2 days from better-positioned alternatives.
- Margin Erosion:** Each additional shipping day costs approximately \$420 in logistics expenses per order, directly compressing gross profit.
- No Decision Support:** There is currently no tool to simulate reassignment scenarios or quantify the impact of changes before execution.

THE APPROACH

How the System Works

- 01

Predictive Modelling A Random Forest model ($R^2=83\%$) was trained on historical orders to predict shipping lead time for any product-factory-region-shipmode combination.
- 02

Scenario Simulation For each of 15 products, all 4 alternative factory options are simulated across 6 regions and 4 ship modes — 1,440 predictions per run.
- 03

Ranked Recommendations Factory reassignments are ranked by a composite score of lead-time reduction (speed) and profit impact, adjustable via a priority slider.

83%	27%	\$5,446	89%
Model Accuracy (Random Forest R^2)	Lead Time Reduction (Top 4 Reassignments)	Estimated Annual Profit Gain	Scenario Confidence Score

TOP RECOMMENDATIONS

Immediate Actions for Leadership

Priority	Product	Move From → To	LT Saving	Profit Gain
URGENT	Fizzy Lifting Drinks	Sugar Shack → Secret Factory	▼ 31.4%	+\$1,904
URGENT	Laffy Taffy	Sugar Shack → Secret Factory	▼ 27.9%	+\$1,410
HIGH	Fun Dip	Sugar Shack → Secret Factory	▼ 26.8%	+\$1,000
HIGH	SweeTARTS	Sugar Shack → The Other Factory	▼ 22.1%	+\$1,132
MEDIUM	Triple Dazzle Caramel	Wicked Choccy's → Lot's O' Nuts	▼ 19.3%	+\$1,020

Recommended Next Step: Pilot the top 2 URGENT reassignments (Fizzy Lifting Drinks and Laffy Taffy) for one quarter. Measure actual vs. predicted lead-time reduction. Use the interactive dashboard's What-If module to confirm factory capacity before execution.

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